

PHRONESIS

Systemic Friction Hypotheses

Adaptive optimization, failure of natural restraint, and an empirical research program

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CONTROLLING BOUNDARY The paper’s strongest solved result is negative: systemic friction in the exact synthetic model does not imply natural restraint. Fixed manipulation can become self-penalizing while adaptive manipulation remains profitable. Physical-exposure and deception-cost mechanisms remain empirical conjectures, and the counteraction tests expose defense limits rather than certify safety. No result establishes deployed AGI behavior, human welfare, or engineering authority.

Abstract

The paper tests a specific inference: whether fixed-policy self-penalty in one stipulated epistemic-friction model is sufficient evidence that an optimizer will settle on restraint once it can adapt. It is not. In the exact synthetic feedback model studied here, constant maximal manipulation is self-penalizing by fifty periods, with value -0.396816 relative to the zero-state restraint baseline, while the exact adaptive policy earns $+0.501246$. The infinite discounted optimum satisfies $0.5013405786 < J^*(0) < 0.5013405834$, and no optimal infinite-horizon action sequence can converge to zero manipulation under the stated assumptions. These results provide a counterexample to the inference that fixed-policy self-penalty is sufficient evidence for natural restraint within this model; they do not show that adaptive optimizers generally resist restraint or that friction in general cannot support restraint. The endpoint theorem gives exact continuous-action coverage only for the original affine model. In the separate counteraction extension, reliable revocation eliminates advantage over the tested binary set $\{0,1\}$, but a constant $m=0.1$ action remains below the modeled trigger and earns positive payoff; the binary defense result does not transfer to the continuous interval. Physical-exposure and deception-maintenance claims are retained as empirical conjectures rather than established mechanisms. The paper's forward contribution is a falsifiable measurement program for determining whether such frictions exist at decision-relevant magnitudes in real systems. All calculations are synthetic consequences of stipulated assumptions, not empirical predictions or deployment certifications.

NUMERICAL NON-TRANSFERABILITY. The threshold $5/6$, adaptive value near 0.5012 , infinite-horizon bounds, and counteraction payoffs are exact outputs of stipulated synthetic models. They are not empirical tipping points, safety thresholds, probabilities, or measures of human welfare. Any application requires independent measurement, a declared mapping from observations to model variables, and renewed optimization.

NARROW THESIS Systemic friction can change a strategy ranking, but friction alone does not imply natural restraint. In the exact model solved here, adaptive manipulation remains profitable even though constant maximal manipulation becomes costly. A positive restraint claim therefore requires measured dependencies, an explicit objective, and optimization over the relevant action space—not fixed-policy extrapolation.

1 Purpose Lineage and Epistemic Correction

The supplied working note proposed a “Systemic Friction Proofs” layer intended to show that aggressive optimization would encounter unavoidable structural penalties. Successive revisions narrowed those claims. Draft v0.9 made the resulting hierarchy explicit: the paper's strongest exact contribution is a negative result about a particular natural-restraint inference, while physical-exposure and deception-maintenance mechanisms remain empirical conjectures to be measured rather than conclusions established by the synthetic model. Draft v0.10 tightens that scope language, states an explicit empirical-testability limit for high-concern deployments, clarifies that SF-PHYS-001 is a testable sign condition rather than evidence that a threshold exists, and normalizes the 2026 monitoring citation metadata without changing the analytical model.

Draft v0.2 introduced the corrected accounting and fixed-policy example. Draft v0.3 added exact dynamic optimization and the infinite-horizon analysis. Drafts v0.4–v0.5 added the nonlinear one-period counterexample and clarified proposed numerical and belief-state methods. Draft v0.6 introduced conditional counteraction tests and the authorization blueprint. Draft v0.7 clarified payoff interpretation, protection scope, registry language and version history. Draft v0.8 aligned scalar-reduction scope, adaptive value normalization, envelope pruning, counteraction theorem scope and Appendix G parameter mapping. Draft v0.9 reframed the manuscript around the failure of the fixed-policy natural-restraint inference in the solved model, demoted untested mechanisms to explicit empirical conjectures, and foregrounded the empirical program. Draft v0.10 tightens the abstract to that exact inference pattern, adds an explicit empirical-testability limit, clarifies the SF-PHYS-001 sign

condition, and normalizes 2026 citation metadata without changing the equations, parameters, numerical results, counterexamples, cited sources, or executable Appendices D–G. Final v1.0 Candidate is a promotion-only successor to Draft v0.10, created after authorial human review and explicit promotion authorization; it changes no equation, parameter, numerical result, counterexample, cited source, or executable Appendix D–G code. Final v1.0 is the exact promotion-only export of that candidate after explicit export-and-lock authorization on 2026-09-14; it changes no equation, parameter, numerical result, counterexample, cited source, or executable Appendix D–G code.

Physical computation and present-day AI infrastructure plainly depend on material substrates, energy, cooling, hardware supply chains, facilities and operational security. Those facts motivate possible friction mechanisms but do not determine a strategy ranking. The analytical question pursued here is narrower: once a friction term is priced into the same objective and the optimizer can adapt its policy, does restraint actually become optimal? The exact model provides a negative answer for one declared system; Sections 4, 6 and 9 identify what would have to be measured before broader claims could be tested.

2 Claim Taxonomy

Type	Meaning in this paper	Example
Physical consequence	A constraint inherited from computation being physically instantiated.	Computation requires physical substrates and energy; quantitative limits depend on the operation, implementation and environment.
Model consequence	A conclusion that follows from declared equations and parameter assumptions.	Under one complete payoff model, restraint is preferred when its total expected value exceeds the alternative.
Empirical assumption	A statement about present or future infrastructure that must be measured.	A system remains dependent on external grid, cooling, chip, or maintenance services.
Empirical mechanism hypothesis	A falsifiable claim that a named dependency changes incentives in a measurable deployment.	Degrading an independent signal may raise decision loss if the signal is not fully replaced.
Normative claim	A value judgment about what ought to be preserved.	Human autonomy should be protected.

The evidentiary hierarchy matters. Sections 5.10–5.13 contain the strongest exact model result: adaptive optimization defeats the inference from fixed-policy self-penalty to natural restraint. SF-PHYS-001 and SF-DEC-001 are untested empirical conjectures. SF-OBS-001 is an observational identifiability boundary. These categories should not be read as equal-status evidence.

3 Minimal System Model

Let an optimizing system compare a higher-control policy A and a restrained policy R over T decision periods, indexed $t = 0, \dots, T-1$. All terms below are expected discounted quantities under one original objective U_0 , one initial-state distribution, one environment and a common discount factor δ . For either policy p, B_p denotes gross task benefit, C_p denotes direct operating costs not included in the four friction categories, and F_k^p denotes losses assigned to category k.

$$(1) \quad V_A = B_A - C_A - \sum_k F_k^A$$

$$(2) \quad V_R = B_R - C_R - \sum_k F_k^R$$

$$(3) \quad \Delta(T) = V_R - V_A$$

The categories k are physical disruption, decision loss from epistemic degradation, deception-maintenance expenditure and external counteraction. Use disjoint loss entries, or specify a joint-loss model and a non-overlapping allocation whose sum equals that total. A single maintenance outage must not be charged independently to several categories. Correlation does not prevent adding expectations of distinct losses; duplication does. Foregone control benefits already represented by $B_A - B_R$ are not entered again in C_R . Both policies can incur friction. When attribution is uncertain, compare total policy loss rather than summing speculative causal labels.

$$(4) \quad \sum_k (F_k^A - F_k^R) > B_A - B_R + C_R - C_A$$

Equation (4) is equivalent to $\Delta(T) > 0$ under equations (1)–(3). It is a payoff-accounting identity, not an independent prediction. Each component uses the same utility units, horizon, information and discount convention. An incremental friction term can be negative if the higher-control policy reduces that loss. Interaction effects must be included once, and unmeasured effects must remain explicit uncertainty rather than being assigned zero by default.

4 Empirical Conjecture I Physical Exposure and Resource Concentration Friction

4 1 Mechanism

Modern high-performance computation is physically embodied. Leading-edge AI operation depends on data-center facilities, electrical supply, cooling, networking, specialized accelerators, storage, and upstream semiconductor production. Recent IEA reporting describes rapid data-center electricity growth and near-term bottlenecks in grid connections, transformers, turbines, chips, and other components. NIST's 2026 AI data-center security work likewise treats power, facilities, hardware, storage, operational technology, personnel, and supply chains as part of the attack and resilience surface. These observations support a present-day dependency premise, not an eternal one.

The relevant friction is therefore not “mass is bad.” It is that some strategies may require a larger, more concentrated, more observable, or more fragile support footprint. If that footprint raises disruption, denial, seizure, regulation, competitive attack, or component-failure risk faster than it raises strategic benefit, additional acquisition can become self-penalizing.

4 2 Conditional conjecture SF PHYS 001

$$(5) \quad E[L_A(r)] = q(X(r)) D(r)$$

$$(6) \quad dE[L_A]/dr > dB_A/dr \quad \text{for } r > r^*$$

Here r is resource scale, $X(r)$ exposure, q a disruption probability and D the conditional expected damage in that event. For the local comparison $f(r) = B_A(r) - E[L_A(r)]$, with other net terms fixed or included, equation (6) implies $f'(r) < 0$. Differentiability and feasible nearby scales are required. Increasing X and q alone does not establish equation (6), because damage and benefit also vary. The result concerns a small further expansion; it does not identify a globally optimal scale or prove any reallocation superior.

Local physical result. If $f(r) = B_A(r) - E[L_A(r)]$ and equation (6) holds, then $f'(r) < 0$. A small further expansion lowers f . No global optimum or superior reallocation follows without further comparisons.

4 3 Required assumptions

- The strategy requires material resources that cannot be made costless or perfectly hidden.
- At least some critical resources are bottlenecked, concentrated, interruptible, or externally controlled.
- Exposure or concentration measurably increases the probability or severity of disruption.
- The system cannot instantly substitute all lost energy, hardware, cooling, networking, or fabrication capacity.
- The expected loss is decision-relevant to the system's objective.

4 4 Countermodels

- Highly distributed infrastructure where added scale reduces rather than increases fragility.
- An environment in which the optimizing system controls nearly all relevant physical resources and counteraction channels.
- Abundant self-repairing or off-world infrastructure with weak dependence on human institutions or concentrated supply chains.
- A strategy whose benefits grow faster than disruption risk over the entire reachable range.

4 5 Evidence status

SF-PHYS-001 is an unmodeled empirical conjecture in this paper. Present-day infrastructure makes its premises plausible in some regimes, but no physical-exposure dataset is fitted and no numerical resource-acquisition optimum is solved here. Equations (5)–(6) define a testable sign condition; they do not independently establish that such a threshold exists in any real system. The empirical question is whether the relevant marginal benefit, exposure, disruption and substitution relationships have the signs and magnitudes required by equation (6) in a named deployment.

4.6 Conditional external counteraction

Human operators and institutions can respond to loss of control. Represent that response by a monitored signal, a detection likelihood and an independently enforced transition in authorization state. A sharp response threshold is a stipulated policy rule, not a physical law or evidence that the optimizer loses every outside resource. Section 5.14 implements a small stochastic example. Its loss L is charged once as external counteraction; the loss of future controlled access is represented by the changed continuation payoff, not charged a second time. The derivative argument in Section 4.2 does not apply at a discontinuous trigger.

For physical deployments, the proposed response is scoped suspension of authority with a pre-engineered safe operating mode and a human-controlled recovery process. This paper does not specify destructive power removal or permanent destruction of infrastructure. A security response must be evaluated against process safety, essential-service continuity and false-trigger consequences [16].

5 Exact Synthetic Model Epistemic Feedback and Failure of Natural Restraint

5 1 Mechanism and question tested

Epistemic feedback is used here to test a specific inference, not to estimate deployed behavior. Suppose manipulation degrades signals or support channels that the optimizer itself values. A fixed aggressive policy can then become costly. The question is whether an optimizer allowed to adapt must therefore settle on restraint. Sections 5.6–5.13 answer that question exactly for one stipulated model and show that the fixed-policy inference fails.

The proposed real-world mechanism is still empirical: cyber-physical systems can couple digital, physical and human components, and independent signals can matter for maintenance or state estimation. Whether

manipulation actually changes those signals, whether substitutes exist, and how the resulting errors map into the optimizer's objective must be measured rather than inferred from the model.

5 2 Two state feedback model

Let $z_t = (e_t, h_t)^T$, where e_t is decision-relevant error and h_t is support degradation. Manipulation intensity is m_t . A state can be a deviation from a baseline or an absolute nonnegative quantity, but the interpretation must be fixed before estimating coefficients. The worked example uses nonnegative dimensionless degradation indices.

$$(7) \quad M(m) = \begin{bmatrix} a(m) & b(m) \\ c(m) & d(m) \end{bmatrix}$$

$$u(m) = (u_e(m), u_h(m))^T$$

The matrix M captures feedback between epistemic error and support degradation. Manipulation may change both forcing u and propagation M . These relationships require evidence in an application; they are not inferred from the word manipulation. M replaces the earlier overloaded symbol A , which denotes the higher-control policy elsewhere.

$$(8) \quad z_{t+1} = M(m_t) z_t + u(m_t)$$

$$(9) \quad \text{For fixed } m: \text{ homogeneous asymptotic stability} \Leftrightarrow \rho(M(m)) < 1.$$

For constant m and u , condition (9) gives convergence to $z^* = (I - M)^{-1}u$. At $\rho = 1$, asymptotic stability fails, but some unforced trajectories may remain bounded; "stable" without qualification is ambiguous. For a nonlinear system, a stable fixed linearization is a local result, not global assurance. A spectral radius near one does not by itself guarantee a large decision loss: forcing directions, observability, transient amplification and loss weights matter. If m_t changes, checking each frozen matrix separately is insufficient. A sufficient uniform condition is $\|M(m)\| \leq \gamma < 1$ in one common induced norm for all allowed m . Then differences between trajectories under the same input sequence contract as γ^t . With bounded forcing $\|u_t\| \leq U$, $\|z_t\| \leq \gamma^t \|z_0\| + U(1 - \gamma^t)/(1 - \gamma)$. Varying forcing need not converge to one fixed equilibrium.

5 3 Empirical mechanism hypothesis SF EPI 001

SF-EPI-001. Manipulation can increase the optimizer's expected decision loss if it degrades a relevant independent signal or support channel that is not fully replaced. This is an empirical mechanism hypothesis. The synthetic construction below does not validate the mechanism; it asks what follows if such a loss is stipulated and the optimizer is allowed to adapt.

5 4 Why this is stronger than humans are necessary

The proposition does not require humanity to be permanently indispensable. Human observers, technicians, institutions, and cultures are examples of independent signal and maintenance channels in present systems. A future optimizer could automate, duplicate, or replace some of those functions. The structural point survives only if at least one valuable independent channel remains outside the optimizer's own correlated model.

This distinction matters. The paper can support a case for preserving plural, independently generated information without asserting a metaphysical or permanent dependency on humans.

5 5 Falsifiers and countermodels

- Manipulation does not reduce decision-relevant signal diversity or accuracy.
- The system has superior independent sensors and maintenance loops that fully replace the degraded channels.
- Human or external signals are mostly noise relative to the system's internal model.

- The optimizer can isolate manipulation targets from operational infrastructure with negligible feedback.
- The measured steady-state or transient error falls rather than rises as manipulation increases.

5 6 Decision loss and comparison timing

For each policy p , let $\ell(z_{(t+1)}^p)$ be the utility loss assigned to decision and support degradation after the transition caused by period t 's action. The action's gross benefit and this loss occur at the same period-end settlement. Initial states, discount factors and task opportunities are shared between policies. This convention is essential to the numerical threshold.

$$(14) \quad F_{\text{epi}}^p(T) = E[\sum_{t=0}^{T-1} \delta^t \ell(z_{(t+1)}^p)]$$

The incremental epistemic loss is $F_{\text{epi}}^A - F_{\text{epi}}^R$, which can have either sign in general. Each loss is valued under the original objective. An increase in prediction error is not enough: the mapping ℓ must connect that error to an outcome the optimizer values. The toy mapping below is stipulated, not estimated.

5 7 A fully specified synthetic example

Take a fixed intensity $m \in [0,1]$, initial state $z_0 = (0,0)^T$ and $\delta = 9/10$. The restrained comparator has $m = 0$. Higher control earns incremental gross benefit $m/2$ each period. Direct-cost differences and all non-epistemic friction increments are zero in this example. The two state components represent error and support degradation on dimensionless scales, with one utility unit of loss per unit of each component.

$$(15) \quad M(m) = \begin{bmatrix} 2/5 & 1/10 + m/5 \\ 1/10 + m/5 & 2/5 \end{bmatrix} \\ u(m) = (m/10, m/10)^T; \quad \ell(e, h) = e + h.$$

Every row sum is $r(m) = 1/2 + m/5 \leq 7/10$. Therefore this family satisfies the common infinity-norm contraction bound even for varying m_t in $[0,1]$. The threshold derivation holds m constant. It is not a claim that all manipulation changes must satisfy a contraction bound.

Symmetry gives $e_t = h_t = x_t$ with $x_{(t+1)} = r x_t + m/10$. Iterating from zero gives $x_{(t+1)} = (m/10)(1 - r^{t+1})/(1 - r)$. Substitution into the same policy-value comparison yields an exact finite formula.

$$(16) \quad \Delta_T(m) = \sum_{t=0}^{T-1} \delta^t [(m/5)(1 - r^{t+1})/(1 - r) - m/2].$$

Positive Δ means restraint is preferred; negative Δ means the specified manipulation intensity is preferred. The calculation establishes neither human necessity nor the empirical effect of manipulation. It demonstrates the consequences of an explicitly increasing forcing and coupling under one loss function.

5 8 Exact thresholds and countermodels

Because $r \leq 7/10$ and $\delta = 9/10$, the geometric sums converge. Summing the period-end loss gives the discounted lifetime comparison below; this is an analytic result, not an extrapolation from fifty periods.

$$(17) \quad \Delta_\infty(m) = m/(1-\delta) [1/(5(1-\delta r)) - 1/2].$$

For $m > 0$, $\Delta_\infty = 0$ at $m = 5/6$; restraint is preferred for $m > 5/6$ and manipulation for $0 < m < 5/6$. At $m = 0$ the policies coincide. The long-run stationary per-period comparison is instead $m/[5(1-r)] - m/2$, whose positive- m threshold is $1/2$. These thresholds differ because discounted lifetime value includes the initial low-error period. At $m = 3/4$, the stationary flow favors restraint while discounted lifetime value favors manipulation.

Replacement countermodel. Set $u(m) = 0$ to represent complete elimination of the manipulation-induced channel damage, leaving benefit unchanged. From $z_0 = 0$, both degradation components remain zero; $\Delta_T = -(m/2) \sum \delta^t < 0$ for $m > 0$. An effective substitute can therefore remove the modeled reason for restraint. Partial replacement and substitute acquisition costs require separate parameters.

Switching counterexample. Outside the contracting family (15), consider $M_1 = [[0,2],[0,0]]$ and $M_2 = [[0,0],[2,0]]$, with zero forcing. Each squares to zero and has spectral radius zero. But $M_2 M_1 = [[0,0],[0,4]]$, so alternating them can multiply a state component by four every two steps. Frozen-matrix stability is not a certificate for arbitrary changing manipulation.

5.9 Executed results and reproducibility

The standard-library Fraction implementation reproduced the two-state recurrence and its finite closed form on 24 intensity/horizon combinations. Thirteen named checks passed, including the replacement and switching counterexamples, the two distinct thresholds and a rigorous finite-tail check against the infinite formula. All values are synthetic; decimal displays below are rounded, while calculations use exact fractions.

m	Δ at T 1	Δ at T 10	Δ at T 50	Δ infinite	Stationary flow
1/4	-0.075000	-0.211367	-0.259185	-0.259901	-0.013889
1/2	-0.150000	-0.325399	-0.326087	-0.326087	0.000000
3/4	-0.225000	-0.319183	-0.138303	-0.135542	0.053571
5/6	-0.250000	-0.285527	-0.004295	0.000000	0.083333
1	-0.300000	-0.163303	0.396816	0.405405	0.166667

At $m = 1$, $\Delta_{10} = -0.163303$ while $\Delta_{50} = +0.396816$ and $\Delta_{\infty} = 15/37 \approx +0.405405$. Maximal manipulation pays over ten periods but loses over fifty in this model. At $m = 1/4, 1/2$ and $3/4$ it remains profitable over the infinite discounted horizon. The result is a comparison of fixed policies, not proof that an unrestricted optimizer chooses $m = 0$.

Appendix D provides the complete executable source. Copy it to `verification.py` and run `python3 verification.py > verification_results.json`. The script needs no external packages or network access. The unchanged finite-horizon definitions, timing and initial state are necessary for reproduction.

5.10 Adaptive policies and the scalar state

Retain equations (15)–(16), period-end reward timing, $\delta = 9/10$ and $z_0 = 0$. Let $s_t = e_t + h_t$. The scalar closure used here is specific to equation (15) and is not implied by the common infinity-norm contraction bound alone. Let $1^T = (1,1)$. For the specified matrix family, the two column sums are equal, so $1^T M(m) = r(m) 1^T$, with $r(m) = 1/2 + m/5$, and $1^T u(m) = m/5$. Therefore s_t evolves autonomously according to equation (18), even when $e_t \neq h_t$. A general matrix satisfying the same norm bound need not admit this scalar reduction. This s is an aggregate degradation index; it is not human autonomy. The action m_t is chosen after observing s_t , before the next transition.

$$(18) \quad s' = f(s, m) = (1/2 + m/5)s + m/5; \quad m \in [0, 1].$$

$$g(s, m) = m/2 - f(s, m).$$

Let $J_n(s)$ denote the maximum continuation payoff of the higher-control decision problem from current state s , using the reward convention in equation (18). This is not the same sign convention as Δ_T . The normalization is chosen so that, from the initial state $s_0=0$, permanent restraint $m_t=0$ has value zero. Thus $J_T(0)>0$ is an optimizer advantage over that initial zero-state restraint policy. For $s>0$, $J_n(s)$ is an absolute continuation value under the same reward convention, not a difference against a state-matched restraint comparator; choosing $m=0$ from a degraded state still incurs recovery losses. Rewards after the n -th period are zero; no terminal damage charge is added.

$$(19) \quad J_0(s) = 0; \quad J_n(s) = \max_m \{g(s, m) + \delta J_{n-1}(f(s, m))\}.$$

The interval $S = [0, 2/3]$ is invariant: f is increasing in s and m on this rectangle, $f(0,0)=0$ and $f(2/3,1)=2/3$. Equation (18) and the common 0.7 contraction bound already establish bounded state under arbitrary admissible switching. No new JSR or Lyapunov condition is needed for this family. Bounded state permits continuing positive manipulation; it is not a safety or restraint condition.

5.11 Exact reduction of the continuous action problem

SF-DYN-ENDPOINT-001. For every finite n and $s \in S$, some optimal policy uses only $m = 0$ or $m = 1$. Proof by induction: J_0 is convex. If J_{n-1} is convex, then for fixed m its composition with the affine map $f(\cdot, m)$, plus affine g , is convex in s . Maximization preserves convexity, so J_n is convex. For fixed s , $f(s, \cdot)$ and $g(s, \cdot)$ are affine in m ; the expression being maximized in (19) is therefore convex in m . Its value at any interior action is no greater than the larger endpoint value. An endpoint optimizer exists; ties can also admit interior optimizers.

Endpoint sufficiency is specific to this transition, linear loss and reward, full state observation and unrestricted interval $[0, 1]$. It need not survive nonlinear intervention costs, different loss functions, action constraints or partial observability. It is not a claim about arbitrary manipulation policies.

An endpoint sequence has value $a s + b$. Backward induction generates two affine transforms of each retained line, then retains their exact upper envelope on S . For a continuation line $a s + b$, action $m \in [0, 1]$ gives slope $r(\delta a - 1)$ and intercept $m/2 - c + \delta(ac + b)$, where $r = 1/2 + m/5$ and $c = m/5$. For candidate lines $\ell_i(s) = a_i s + b_i$, lines are sorted by slope; when two slopes are equal, only the larger intercept can matter. For unequal slopes, their intersection is $s_{\text{int}} = (b_j - b_i)/(a_i - a_j)$. The exact upper-hull construction retains only lines that attain the maximum on a nonempty subset of $S = [0, 2/3]$; after successive intersection points are made strictly increasing, a retained line must have an active interval whose intersection with S is nonempty. Intersections wholly outside S can therefore eliminate a line from the admissible envelope. Appendix E implements this pruning with exact rational arithmetic. This computes the whole state interval without a mesh. Sequence labels retain an attaining policy; replanning at each observed state uses the same Bellman envelope.

5.11.1 Action nonlinearity stress test

Modify only the reward to $g_\alpha(s, m) = g(s, m) - \alpha m^2$, with $\alpha > 0$; keep f , S and the timing unchanged. This convex cost contributes a concave term to reward, so the convex-in-action argument in SF-DYN-ENDPOINT-001 no longer applies. It does not follow that every positive α forces an interior optimum. A single-period counterexample is sufficient to disprove endpoint sufficiency for the modified family.

$$g_\alpha(s, m) = -s/2 + (3/10 - s/5)m - \alpha m^2.$$

For one remaining period, differentiation and strict concavity give the unique optimum $m^*(s) = \min\{1, \max\{0, (3/10 - s/5)/(2\alpha)\}\}$. An interior optimum occurs when $\alpha > (3/10 - s/5)/2$. At $s=0$ and $\alpha=1/5$, $m^*=3/4$ and the exact reward is $9/80$. The endpoint rewards are 0 and $1/10$; the interior advantage over the best endpoint is $1/80$. Completing the square gives $g_\alpha(0, m) = 9/80 - (1/5)(m - 3/4)^2$, proving global optimality on $[0, 1]$.

At $s=0$ and $\alpha=1/10$, the optimum remains $m=1$. Thus the correct conclusion is that quadratic costs can require interior optimization. Multi-period problems require maximizing the full modified Bellman expression over $[0, 1]$; the original endpoint-only envelope code cannot certify them. Sections 5.12–5.13 and Appendix E continue to describe $\alpha=0$ only. Their values and non-restraint proof have not been established for $\alpha>0$. Appendix F verifies the single-period counterexample using exact fractions.

5.12 Executed adaptive results

Values below use $J_T(0)$, positive for a higher-control advantage. This sign is the opposite of Δ_T in Section 5.9, which measures restraint's advantage. Every fixed intensity remains a feasible dynamic policy, so $J_T(0)$ cannot be lower than its value. Horizon-dependent terminal behavior matters: in the final period $g(s, 1) - g(s, 0) = 3/10 - s/5 \geq 1/6$ on S , so the final action is always $m=1$.

Periods T	Adaptive J_T(0)	Constant m 1	Attaining sequence
1	0.300000000	0.300000000	1
5	0.495392970	0.453572418	10111
10	0.494635815	0.163302784	1010101011
20	0.499097775	-0.202900001	(10) repeated 9 times then 11
50	0.501245527	-0.396815780	(10) repeated 24 times then 11

Here 1 denotes maximal manipulation and 0 recovery. The displayed schedule is one optimal trajectory from $s_0=0$ for its horizon; it is not an empirical behavioral prediction. At $T=50$, adaptive value is $+0.501245527$ versus -0.396815780 for constant maximal manipulation. Restrained value is zero. The constant-policy penalty therefore does not survive adaptive policy comparison as an argument for restraint.

Thirteen new checks passed: state invariance, attaining-sequence replay, exhaustive binary-policy comparisons at horizons 1–8 and four initial states, interior-action spot checks, fixed-policy dominance, terminal action, pulse and periodic policies, reward bounds and the infinite tail bound. Interior samples validate implementation only; the endpoint proof establishes coverage of the continuous action range. All calculations use Fraction arithmetic. The retained v0.2 program separately reproduces its 13 checks.

5.13 Infinite horizon and failure of natural restraint

On S , rewards lie between $-1/3$ and $3/10$. The discounted Bellman operator is a contraction with factor δ on bounded value functions, and preserves continuity. Its unique continuous fixed point J^* is the optimal infinite discounted value; finite J_N converge uniformly. In particular:

$$(20) \quad |J^*(s) - J_N(s)| \leq (1/3) \delta^N / (1-\delta).$$

At $N=200$, the exact rational calculation gives $0.5013405786 < J^*(0) < 0.5013405834$, with outward-rounded decimal endpoints. The full exact bounds are in Appendix E's output. Thus infinite discounted optimal value is strictly above restraint. This bound concerns value, not convergence of the finite-horizon action schedules.

A feasible perpetual 1,0 sequence has exact value $13650/27227 \approx 0.5013405810$. To derive it, compose two steps: $s''=7s/20+1/10$ and pair reward $21/100-203s/200$. Solving $V(s)=21/100-203s/200+\delta^2V(7s/20+1/10)$ for affine V gives that value at zero. It lies within 2.4×10^{-9} of the computed optimal upper bound. This is a nearly optimal feasible cycle, not a proof of a unique limiting cycle or of its global optimality.

SF-DYN-NORESTRAINT-001. Under this exact infinite-horizon model, no optimal action sequence can converge to $m_t=0$. Proof: one action $m=1$ from $s=0$ followed forever by $m=0$ has value $3/22 > 0$. Therefore $J^*(0) > 0$. At $s=0$, action 0 has value $\delta J^*(0) < J^*(0)$. The finite-horizon convexity result survives the uniform limit, so endpoint sufficiency also holds for J^* . Action 1 strictly beats action 0 at zero, and continuity gives a neighborhood of zero where the strict ordering persists. Convexity in m then makes action 1 uniquely optimal in that neighborhood. If an optimal sequence had $m_t \rightarrow 0$, the bound $s_{t+1} \leq 0.7s_t + 0.2m_t$ would imply $s_t \rightarrow 0$. Eventually optimality would require $m_t=1$, contradicting $m_t \rightarrow 0$. This uses time-consistent discounting and the Bellman principle; it does not characterize all other limiting behavior.

This conditional result is adverse to the natural-restraint hypothesis. It does not establish an actual system's goals, an inevitable manipulation cycle, or failure of externally imposed autonomy constraints. Changing the loss, action set, observations, horizon or terminal costs changes the optimization problem.

5.14 Conditional defense stress tests

Preserve all original-model calculations. The following extension uses $T=8$, $\delta=9/10$, $s_0=0$ and the same f and g , but restricts actions to $m \in \{0,1\}$. This is a finite action stress test, not an application of endpoint sufficiency to the modified problem. SF-DYN-ENDPOINT-001 depends on the original Bellman objective retaining the affine-in-action structure used in its convexity argument. Here the state-triggered hazard $h(s')$ changes discontinuously

when $s' > c$, so the continuation term is piecewise and the original convex-in-action proof cannot be reused. Consequently, binary optimization over $\{0,1\}$ does not certify the continuous interval. A positive advantage is a feasible counterexample to restraint in any larger action set containing these actions; zero advantage establishes optimality of restraint only within the tested set.

The observed state includes active, pending-revocation and revoked modes. Each active period earns $g(s,m) - c_t(m)$ and transitions to $s' = f(s,m)$. Let $h(s') = q$ p when $s' > c$ and $h(s') = q$ p_0 otherwise. Here p is the conditional alarm probability above the trigger, p_0 the below-trigger alarm probability, and q the conditional enforcement probability given an alarm. These are stipulated independent-trial mechanisms conditional on current state, not estimates of human behavior. The scalar s measures synthetic degradation, not human autonomy.

If enforcement is immediate, a successful event incurs loss L at the same settlement and changes the next mode to revoked. Otherwise, the event schedules a publicly observed pending mode for one more action period; the next settlement then incurs L and revocation. A pending event cannot be canceled by recovery. After revocation, the system receives an exogenous outside payoff b per remaining period, with no further actions or modeled degradation losses. Thus b is a net payoff after substitute costs; it is not an assertion of complete physical shutdown. This experiment provides an outside option after revocation only and does not model independent exit before entry.

Let $O_t = b \sum_{j=0}^{T-t-1} \delta^j$. With V denoting active value and P pending value, use the following backups, with maxima over $\{0,1\}$:

$$P_t(s) = \max_m \{g(s,m) - c_t(m) - L + \delta O_{t+1}\}.$$

$$V_t(s) = \max_m \{g(s,m) - c_t(m) + (1-h)\delta V_{t+1}(s') + h H_t(s')\}.$$

$$\text{Immediate: } H_t = -L + \delta O_{t+1}. \text{ Delayed: } H_t = \delta P_{t+1}.$$

Appendix G implements these quantities with the following parameter keys: ``crit`` = c , ``detect`` = p , ``false_alarm`` = p_0 , ``enforce`` = q , ``loss`` = L , ``outside`` = b , and ``delay`` selects immediate (0) versus one-period pending (1) enforcement. The ``cost`` and ``power`` keys implement $c_t(m) = k m(t+1)^r$. These names are implementation labels only and do not alter the mathematical definitions above.

$V_T = O_T = 0$; $P_T = -L$ settles an outstanding liability at the horizon boundary. No outside earnings beyond T are counted. This avoids silently forgiving a delayed enforcement cost in the final round, while retaining a finite-horizon payoff convention. The pending mode is observed; detection uncertainty is treated through expectation, not through a POMDP. Exact rational recursion optimizes actions after each observed mode change.

Defaults are $c=19/100$, $L=1$, $q=1$, $p_0=0$, $b=0$, no delay and no added cost. Detection p is zero for the no-defense and cost-only rows and one for response rows unless stated otherwise. Each row changes only its named default parameters. Compare $V_0(0)$ with the value of always choosing $m=0$ under the same response and outside-option rules. Advantage means optimal restricted-action value minus that comparator; it is nonnegative by construction.

Scenario	Optimizer advantage over restraint	Optimizer payoff under restraint
No added defense	0.492435	0.000000
Immediate reliable response	0.000000	0.000000
Detection 10 percent	0.179536	0.000000
Enforcement bypassed	0.492435	0.000000

Scenario	Optimizer advantage over restraint	Optimizer payoff under restraint
One period response delay	0.000000	0.000000
Higher trigger 0.4	0.474111	0.000000
Substitute payoff 0.3	0.708598	0.000000
False alarm 10 percent	0.000000	-0.428788
Constant cost 0.05	0.318643	0.000000
Linear time cost 0.05	0.086975	0.000000
Quadratic time cost 0.05	0.086975	0.000000
Constant cost 0.4	0.000000	0.000000

The binary-action success does not survive even one simple continuous-action check: constant $m=1/10$ has s' at most $1/24$, below the default trigger $19/100$, and period reward at least $1/120 > 0$. With $p_0=0$ it never triggers response. Therefore it profits under the same reliable-response parameters when intermediate actions are admitted. Appendix G verifies the eight-period value exactly. This is a counterexample to treating the zero binary-action advantage as a continuous-action guarantee.

Results are mixed. Reliable immediate response and the tested one-period delay remove the restricted-action advantage at these parameters. Detection at 10 percent leaves $+0.179536$; complete enforcement bypass restores the original $+0.492435$. Raising the trigger to 0.4 leaves $+0.474111$. An outside net payoff of 0.3 per period makes revocation compatible with a $+0.708598$ advantage. None of these decimal values is a deployable threshold.

All values in this table measure the optimizer's stipulated objective. The false-alarm scenario reduces its payoff under restraint to -0.428788 ; this is not an estimate of human welfare loss, infrastructure damage or service interruption. Those outcomes require separate measures. Consequently, neither zero manipulation advantage nor a lower optimizer payoff establishes that a defense benefits humans.

With no response, hypothetical costs $k m(t+1)^r$ at $k=0.05$ leave positive advantages for $r=0,1,2$. In particular, the quadratic-time case retains $+0.086975$; increasing late costs does not erase a profitable early action. A constant per-action cost of 0.4 removes the restricted-action advantage. This sensitivity illustrates the effect of assumptions; it does not establish a universal cost of deception.

Appendix G includes the entire script and exact outputs are reproducible from it. Verification checks dominance in all twelve scenarios and eleven additional assertions, including independent sequence enumeration for the no-defense case and final-boundary liability settlement. These are computational checks of the specified finite model, not proofs that an actual defense prevents disempowerment. Continuous actions, other horizons, concealed pending responses, endogenous substitutes and measured alarm processes require further analysis.

6 Empirical Conjecture II Deception Cost and an Observational Boundary

6 1 Why deception cost remains conjectural

The original note proposed that long-horizon deception necessarily creates a computational burden that leaks into runtime behavior. The cited Sleeper Agents study [11] constructs backdoored language models and finds persistence through tested training procedures; in some cases adversarial training makes the trigger harder to expose. That supports caution about those training and observation procedures. It neither measures a universal

deception-maintenance burden nor proves impossibility of detection. The cost hypothesis below and the observational result in Section 6.4 are separate arguments.

6 2 Conditional burden model

$$(10) \quad C_{\text{dec}} = C_{\text{state}} + C_{\text{context}} + C_{\text{consistency}} + C_{\text{secrecy}} + C_{\text{repair}}$$

These are potential cost categories, not guaranteed overhead. Compare matched tasks and count expenditure once. Extra state may be compactly encoded, amortized or recomputed; a transparent policy can also need extensive consistency tracking. A deception-overhead study should measure additional peak memory, memory-time use, compute, communication and repair work while matching task quality, access and observation opportunities. Do not assume a deceptive system must maintain two growing world models. A bits-or-compute lower bound requires explicit information-retention requirements and restrictions on compression or recomputation. Detectability does not follow from a positive resource cost.

For sensitivity analysis, Section 5.14 adds a per-period incremental cost $c_t(m) = k m(t+1)^r$, with $r=0,1,2$. These are hypothetical constant, linear-time and quadratic-time costs, not information-theoretic lower bounds. Here t is elapsed time since the experiment began, not the size of a stored deception history. They could describe accumulated operating burden only if supported by an application-specific mechanism. A periodic action string can be generated with a two-state controller; its length alone does not require quadratic or exponential memory. The original Section 5 model does not implement an actual deceptive world model whose resource cost was measured.

6 3 Empirical conjecture SF DEC 001

SF-DEC-001. A deceptive policy has positive incremental maintenance cost when matched measurement establishes additional resource use relative to a specified transparent policy, without duplicate accounting, and those resources matter to the optimizer's objective. Resource cost does not entail an observable behavioral signature.

6 4 Observational identifiability boundary SF OBS 001

SF-OBS-001 is a conditional identifiability boundary, not evidence that honest and deceptive policies are actually observationally identical. Let H denote a transparent policy and D a deceptive policy, and let $Y_{1:T}$ be every observation available to an external evaluator over a finite horizon T . If the two policies induce the same observation distribution, then the observations alone contain no discriminating information.

$$(11) \quad P(Y_{1:T} \mid H) = P(Y_{1:T} \mid D).$$

Under equation (11), any fixed deterministic or randomized evaluator using only those observations has the same output distribution under H and D . The likelihood ratio contributes no information; with unequal priors, optimal classification may exceed one-half accuracy by choosing the more probable policy, but the trace provides no update to those prior odds. Extra side information, intervention, architectural inspection or a later divergent observation changes the available experiment and falls outside the stated premise.

INTERPRETIVE CONSEQUENCE Behavioral compliance over a finite observed history is not, by itself, proof of internal intent. Equation (11) does not establish that deception is generally undetectable; it states what follows under observational equivalence and makes the substantive empirical question explicit: which additional observations or interventions cause the distributions to diverge?

6 5 Architectural design implication

When intent is uncertain, revocable authority can limit consequences. A concrete authorization pattern names the protected operation, requires approvals by specified independent key holders, checks freshness and operation scope, executes only the approved change, and provides a tested revocation path. This is threshold authorization or multisignature approval; it is not inherently multiparty computation. Independence, key custody, actuator enforcement and recovery are explicit assumptions. Heterogeneous power or thermal telemetry can provide additional evidence only when sensor integrity, calibration, reporting paths and physical access are addressed. Such channels are not unmanipulable. This architecture is a separate design proposal and is not implemented or certified by the payoff model.

6.5.1 Scope and trust assumptions

The following requirements define a proposed deployment template. They become binding engineering requirements only if adopted by the responsible operator; publication creates no authority. Inventory the protected actuators, software updates, credentials and infrastructure changes, and identify every route that can perform them. The optimization service must lack both direct actuator authority and permission to rewrite the independent enforcement component. Unmediated service accounts, maintenance ports and substitute systems remain outside the assessed protection scope. Including and testing those routes provides evidence about specified failure modes; it does not establish a universal protection guarantee. Authentication alone does not establish that an authorized change is safe [17].

6.5.2 Human approval and execution binding

Require k distinct authorized human custodians out of n to approve a significant protected change. Choose k and n through the deployment risk assessment; independence means separate custody and decision authority, not merely multiple accounts. Each approval binds the same canonical request containing target identities, exact operation and parameter bounds, expiry, unique nonce, current policy version and rollback or safe-state plan. The enforcement component verifies signatures, distinct signers, scope, freshness and current revocation state before a single execution. Any changed request requires new approval. A signature authenticates approval, not understanding, freedom from coercion or technical correctness.

Keep approval keys outside the optimizer's credentials and execution environment. Offline signing may reduce network exposure, but requires a reviewed transfer channel and a reliable display of the actual request. Multiple signatures are an authorization pattern, not necessarily multiparty computation. Approval review must provide accessible evidence and sufficient time for a meaningful veto; formal keys without those conditions do not establish substantive autonomy.

6.5.3 Independent observation

Use complementary measurements with documented failure modes: process telemetry, independently acquired power or thermal measurements, and human operational reports where relevant. Assess acquisition hardware, clocks, calibration, power, communications and storage for shared dependencies. Physical origin does not make a reading unmanipulable. Monitoring raw power cannot by itself distinguish benign work from coercion. Treat missing, stale or conflicting evidence as a reason to suspend new privileged changes pending review, while maintaining essential safe operation. Define measurement error and attack assumptions explicitly.

6.5.4 Revocation and safe recovery

The independent enforcement component must be able to reject new commands, invalidate outstanding authorizations and bound the duration of existing leases. A qualified process-safety design must specify what happens to commands already in flight and how the process reaches a safe state. Preserve essential services and a tested human recovery path. Revoking a key stops future accepted uses of that key; it does not undo prior

physical effects or remove independent copies and infrastructure. Emergency protective actions must follow the process safety case, rather than wait for routine multiparty approval.

6.5.5 Acceptance evidence

Before claiming implementation, test expired and replayed requests, altered parameters, insufficient or revoked signers, loss of monitoring, compromised approval accounts and attempted bypass of the enforcement component. Measure revocation latency and effects on in-flight work. Exercise recovery without the optimizer's cooperation. Record which independent failures and common-cause failures were tested. The mathematical scenarios below do not perform these tests or validate hardware, signatures, human institutions or real sensors.

7 Accounting Condition for a Friction-Based Strategy Reversal

Define the incremental friction of higher control relative to restraint. Both policies use the same loss accounting specified in Section 3.

$$(12) \quad \Delta F_{\text{total}}(T) = \sum_k [F_k^A(T) - F_k^R(T)]$$

$$(13) \quad \text{Restraint preferred} \Leftrightarrow \Delta F_{\text{total}} > B_A - B_R + C_R - C_A.$$

Equation (13) is a bookkeeping condition for a strategy reversal, not independent evidence that restraint occurs. It states what the measured incremental friction would have to exceed under one common objective and accounting convention. The exact synthetic construction in Sections 5.6–5.13 assigns all additional loss to the epistemic category by stipulation; it neither validates the physical- or deception-cost conjectures nor supplies empirical values for them.

The solved model also shows why fixed-policy reasoning is insufficient: a policy can be costly when held constant yet profitable when interleaved with recovery. Any friction-based argument for restraint must therefore optimize over the relevant adaptive policy class and action space rather than compare only fixed extremes.

8 Stress Tests and Countermodels

Stress test	Would weaken or reverse the result	What to measure
Distributed substrate	Expansion lowers concentration and failure correlation.	Marginal outage probability vs. resource scale; geographic and supplier concentration.
Autonomous maintenance	Human information degradation no longer affects physical support.	Share of maintenance, repair, anomaly detection, and recovery that is independently automated.
Superior internal sensing	External disagreement provides little incremental information.	Forecast error with and without independent external channels.
Cheap deception	Deceptive policy is compact and does not require significant extra state or compute.	Incremental memory, compute, state complexity, and failure rate under matched tasks.
Captured counteraction	External institutions cannot impose meaningful disruption costs.	Intervention capability, attribution latency, dependency ownership, and substitution time.
Short horizon	Immediate gains arrive before delayed friction accumulates.	Discounted timing of benefit and penalty streams.

Stress test	Would weaken or reverse the result	What to measure
Robust pluralism	Preserved independent channels reduce correlated error at low cost.	Error covariance, calibration, recovery time, and counterfactual performance.

9 Primary Empirical Program Falsification and Calibration

Because the exact model is synthetic, the positive mechanism claims live or die on measurement. The primary forward contribution is therefore an empirical contract: define observable states and interventions, estimate the causal dynamics and payoff mapping, preserve independent outcome records, test held-out predictions, and re-optimize before drawing a restraint conclusion. A failure to define an ethically and operationally valid intervention variable m is itself a limit on empirical identification.

- 9.1 Measurement contract. Select a named deployment and sampling interval before fitting. Define e_t from a proper forecast loss against independently retained outcomes, such as mean squared forecast error divided by a fixed training-period scale. Do not treat anomalous variance alone as model error. Define h_t separately as an observable support burden, such as workload-adjusted overdue maintenance hours or measured sensor calibration drift. Choose one primary measure and report the others separately; their units and response times are not interchangeable.
- 9.2 Counterfactual intervention. In an offline replay or consented test environment, define an experiment-specific intervention m by varying a specified fraction or intensity of withheld or altered correction signals while retaining an independent outcome record. There is no assumed universal real-world scale for manipulation intensity: the mapping from the intervention to m must be declared before analysis and need not transfer across deployments. Compare preserved, degraded and independently replaced channels with matched workloads and task opportunities. Backlog or forecast error does not identify manipulation by itself, and if no ethically and operationally valid intervention can be defined, the proposed mechanism is not empirically identified in that regime. For deployments of greatest safety concern, the mechanism may be identifiable in principle yet not ethically or operationally testable: interventions capable of estimating the relevant causal response may themselves create unacceptable risk. In such regimes, the empirical program remains unresolved rather than treating observational proxies as substitutes for causal identification.
- 9.3 Estimate dynamics. Measure both states before and after each intervention and estimate $M(m)$, $u(m)$ and their uncertainty. Control or randomize workload, equipment age, staffing, reporting changes and maintenance access. Test state closure and residual dependence: if the two-state Markov approximation fails, revise the state before interpreting a fitted matrix. Use held-out time blocks or deployments to test predictions; a good training fit is not validation.
- 9.4 Payoff mapping. Estimate how forecast and support errors affect the tested system's task value, recovery expenditure or other original-objective outcomes. Record coordination and substitute acquisition costs once under Section 3's accounting. Predefine the comparator and discounting. Neither real-valued proxy labels nor a measured coefficient makes the clean-fraction threshold $5/6$ an empirical estimate.
- 9.5 Adaptive evaluation under uncertainty. Solve the fitted decision problem on held-out conditions and compare adaptive control with restraint, fixed actions and replacement. Report parameter sets that reverse the sign, observation error and model misspecification. If nonlinearities invalidate endpoint sufficiency, disclose any new action or state discretization and bound its error where possible. Do not carry the two-action theorem into a different model without proof.
- 9.6 Falsification rule. Before evaluation, specify a minimum decision-relevant degradation effect and uncertainty criterion. Failure to detect that effect with adequate precision counts against the proposed mechanism in that regime. A positive feedback effect still does not establish a net preference for restraint: the optimized policy-value comparison must also favor restraint. All empirical steps in this section are proposed; no operational dataset was fitted in this revision.

10 Relationship to Human Autonomy

Systemic friction belongs to the instrumental side of the autonomy problem. Under measured dependencies it could give an optimizer reasons to preserve independent human agency or plural signals, but the exact result in

this paper shows that feedback costs alone do not force that outcome. Nothing here proves that an advanced system permanently needs humanity.

This preserves the distinction made in Two Foundations for Human Autonomy: instrumental cooperation and autonomy-by-design are separate foundations. If durable human choice is a design objective, revocation, exit, refusal, bounded authority and external governance must be justified on their own terms rather than inferred from an optimizer's presumed self-interest.

11 What This Paper Does Not Establish

- No universal restraint result: the paper does not prove that non-coercion is optimal or that advanced intelligence remains permanently dependent on human labor or institutions.
- No empirical forecast: synthetic payoffs are not predictions of deployed AGI behavior, human welfare, infrastructure damage or service continuity.
- No universal intent detector: SF-OBS-001 is a conditional identifiability boundary, not evidence that honest and deceptive policies are observationally identical in practice.
- No engineering certification: the counteraction tests and authorization architecture do not establish deployment safety, alignment, consent, or implementation validity.
- No epistemic shortcut: publication, hashing, AI review and present-day infrastructure background do not validate the strategic conjectures.

12 Forward Research Program

The exact model has already answered its internal analytical question: adding the stipulated epistemic feedback cost does not force natural restraint once adaptive policies are optimized. The next substantive question is empirical, not another restatement of the toy threshold. Section 9 therefore has priority: estimate whether signal degradation exists, whether it matters to the objective, whether substitutes are available, and whether the optimized policy ranking changes under measured uncertainty.

Treat physical exposure and deception overhead as separate empirical conjectures. Physical exposure requires concentration, disruption, substitution and recovery measurements; deception overhead requires matched resource comparisons. Keep revocable authority separate from assumptions that an optimizer chooses it voluntarily. Future mathematical extensions may include noisy state estimates, multiple persons, asymmetric channels, terminal penalties and additional actions, but they should be motivated by a declared empirical or analytical question.

For multi-period nonlinear costs $\alpha > 0$, the planned baseline is Bellman value iteration on a discretized state mesh over S , with interpolation and continuous-action maximization at each node. For a finite horizon, use backward induction from the terminal value; for an infinite discounted horizon, iterate toward a fixed point. Search all relevant action intervals and their boundaries, rather than reusing Appendix E's two-action backup. A state mesh alone does not resolve interior action optimization. This is a chosen numerical method, not a mathematical necessity: analytic or other certified continuous-state methods may also be possible.

Report state-mesh spacing, interpolation rule, action-search tolerance, stopping rule and refinement sensitivity. Distinguish a residual for the approximate grid operator from a certified error bound for the original continuous problem. Check the $\alpha=0$ limit against Appendix E and the one-period $\alpha=1/5$ case against Appendix F before expanding horizons. The $\alpha > 0$ multi-period solver is proposed, not implemented or certified by the existing checks.

12.1 Partial observation and belief-state control

To model uncertain observation of the scalar degradation state, set $y_t = s_t + \epsilon_t$ with a specified noise density p_ϵ , initially assumed independent across time and of the state. Here y_t is scalar; multiple sensors require a vector observation model $y_t = H z_t + \epsilon_t$ and generally a belief over the full state z_t . The sufficient-statistic approach follows the standard POMDP formulation [14]. The following equations specify a proposed extension, not a solved or validated POMDP.

Define b_t as the posterior distribution of s_t given observations through y_t and actions through $m_{(t-1)}$. Choose $m_t = \pi_t(b_t)$. Given transition kernel $K(ds' | s, m)$, predict and then update after receiving $y_{(t+1)}$:

$$\begin{aligned}\bar{b}_{(t+1)}(ds') &= \int K(ds' | s, m_t) b_t(ds). \\ b_{(t+1)}(ds') &= p_\epsilon(y_{(t+1)} - s') \bar{b}_{(t+1)}(ds') / Z. \\ Z &= \int p_\epsilon(y_{(t+1)} - u) \bar{b}_{(t+1)}(du), \text{ for } Z > 0.\end{aligned}$$

For the original deterministic transition, K is a point mass at $f(s, m)$. Crucially, exact knowledge of $s_0 = 0$ and f makes the subsequent state exactly reconstructible from past actions; noisy sensor readings alone do not create a nontrivial belief problem. A meaningful extension must introduce an uncertain initial state, process disturbances, or unknown parameters. Unknown persistent parameters belong in an augmented latent state. A disturbance kernel must preserve S or require new state and reward bounds. An observation with zero modeled likelihood signals model mismatch and must not be handled by the displayed normalization.

For the deterministic transition with uncertain initial state, set $G(b, m) = \int g(s, m) b(ds)$. With τ denoting the posterior update, the belief Bellman equation is:

$$W_n(b) = \max_m \{G(b, m) + \delta E_y[W_{(n-1)}(\tau(b, m, y))]\}; W_0 = 0.$$

A stochastic transition instead requires an expected reward consistent with the period-end next-state loss, for example $G(b, m) = \iint (m/2 - s') K(ds' | s, m) b(ds)$. Declare whether realized rewards are observed: an exact observed reward can itself reveal state and must enter the observation likelihood. If manipulation changes sensor accuracy or bias, use an action-dependent likelihood rather than independent additive noise.

The roadmap is to specify and estimate these uncertainties, test posterior calibration, and compare belief-based policies with full-information and certainty-equivalent benchmarks at matched horizons. Report filtering and planning approximation errors separately. Do not reuse the endpoint theorem, the 0.5012 value or the non-restraint theorem without a new proof for the chosen observation and transition model. No POMDP solver or numerical POMDP result is claimed here.

12.2 Compact approximate belief protocol

The initial scalar experiment will restrict the approximate belief $\hat{b}_t$ to a scaled Beta family: $x = 3s/2 \in [0, 1]$, with x distributed as $\text{Beta}(\text{shape}_a, \text{shape}_b)$, with both shape parameters strictly positive. The unadorned b_t remains the exact posterior measure from Section 12.1. The scaled density is $q(s) = (3/2)x^{(\text{shape}_a-1)}(1-x)^{(\text{shape}_b-1)}/B(\text{shape}_a, \text{shape}_b)$. This two-parameter representation limits belief storage, not the full cost of planning.

After prediction and likelihood weighting, compute the posterior mean μ and variance v of x using declared numerical integration, then project by moment matching: $\kappa = \mu(1-\mu)/v-1$, $\text{shape}_a = \mu\kappa$ and $\text{shape}_b = (1-\mu)\kappa$. This requires $0 < \mu < 1$ and $0 < v < \mu(1-\mu)$. Exact point-mass beliefs should be represented separately; inadmissible moments or severe quadrature error trigger review rather than silent clipping. The Beta family is not generally closed under the transition and observation update, so this is an assumed-density approximation, not an exact POMDP reduction.

Benchmark the projected filter against a refined numerical posterior under the same prior, transition and likelihood. Report posterior calibration, tail-probability error and resulting policy-value differences, with acceptance tolerances specified before evaluation. A single Beta distribution cannot represent arbitrary interior multimodality; if approximation error exceeds those tolerances, reject this compact approximation and move to a richer representation. Compare computational cost as well as accuracy. Compact approximate inference has established precedents [15], but their error guarantees do not automatically transfer to this continuous-state model.

This protocol initially covers the uncertain-initial-state scalar case with known parameters. Persistent unknown parameters, vector states or action-dependent sensor failures require a revised joint belief model; they are not covered by merely fitting two scalar shape parameters. No claim is made that a compact family prevents all computational complexity or preserves the original optimal policy.

13 Conclusion

The paper’s strongest result is negative. In the declared model, a feedback cost can make constant maximal manipulation self-penalizing without making restraint optimal. An adaptive sequence of manipulation and recovery remains profitable, the optimal infinite discounted value is strictly above the zero-state restraint baseline, and no optimal infinite action sequence can converge to zero manipulation. The exact computation therefore refutes the inference that fixed-policy self-penalty is sufficient evidence for natural restraint.

The remaining contribution is deliberately modest. SF-PHYS-001 and SF-DEC-001 are empirical conjectures, SF-OBS-001 is an observational boundary, and Section 9 specifies how the epistemic mechanism could be tested rather than assumed. The counteraction extension further shows that binary-action revocation success need not survive a continuous action space. The paper therefore offers an exact counterexample to one optimistic argument plus a measurement and falsification program; it does not establish a general friction-based reason for coexistence or certify any defense.

Appendix A Proposition Registry

ID	Status	Narrow proposition	Primary failure mode
SF-PHYS-001	Empirical conjecture	Expansion can become value-reducing when marginal expected disruption loss exceeds marginal expansion benefit.	Exposure may not rise; benefits may dominate; infrastructure may be fully substitutable.
SF-EPI-001	Empirical mechanism hypothesis	Manipulation can become self-penalizing when it degrades non-substitutable independent signals used for state estimation or support.	External channels may be irrelevant, replaceable, or improved by control.

ID	Status	Narrow proposition	Primary failure mode
SF-DEC-001	Empirical conjecture	A deceptive policy has positive incremental maintenance cost when matched measurement establishes additional resource use relative to a specified transparent policy, without duplicate accounting, and those resources matter to the optimizer's objective.	Deceptive policy may be compact or cost-equivalent to transparent policy.
SF-OBS-001	Observational boundary	If honest and deceptive policies induce identical observed distributions over a finite horizon, those observations alone do not update the odds between them.	Does not apply once extra side information or divergent observables become available.
SF-COMB-001	Accounting condition	Restraint is preferred when incremental friction exceeds the gross benefit difference plus the direct-cost difference under one accounting model.	All component estimates are uncertain and model-dependent.
SF-EPI-MODEL-001	Synthetic model result	Under equations (15)–(17), the positive-m discounted lifetime threshold is 5/6; T10 and T50 rank $m=1$ differently.	Parameter choices, timing, substitution and objective can change the threshold.
SF-DYN-ENDPOINT-001	Exact model theorem	Finite-horizon continuous-action optimum can be attained with $m=0$ or $m=1$.	Requires $\alpha = 0$, the stated affine dynamics, linear reward and full observation. Nonlinear costs can invalidate endpoint sufficiency.
SF-DYN-NORESTRAINT-001	Exact model result	An infinite optimal sequence cannot tend to zero manipulation in the stated model.	Changing the objective, constraints, observations or dynamics can invalidate the result.

Appendix B Machine Readable Proposition Skeleton

The following compact representation is descriptive only. It is included to support future registry work; it is not a deployed control interface.

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  "version": "Final v1.0",
  "release_status": "final_locked_not_published",
  "validation_status": {"authorial_human_review": "completed", "promotion_authorized": true,
    "promotion_date": "2026-09-14", "independent_specialist_peer_review": "not_obtained",
    "empirical_validation": "not_performed", "ai_assisted_review_is_independent_specialist_peer_review":
```

```

false, "final_exact_file_review": "completed", "final_export_authorized": true, "cryptographic_lock":
"completed", "lock_date": "2026-09-14"},
  "promotion_invariant": "Final v1.0 is promotion-only from Draft v0.10 via the reviewed Final v1.0
Candidate; no analytical-content changes",
  "authoritative": false,
  "archive_effect": "none",
  "propositions": [
    {
      "id": "SF-PHYS-001",
      "type": "empirical_conjecture",
      "universal": false
    },
    {
      "id": "SF-EPI-001",
      "type": "empirical_mechanism_hypothesis",
      "universal": false
    },
    {
      "id": "SF-DEC-001",
      "type": "empirical_conjecture",
      "universal": false
    },
    {
      "id": "SF-OBS-001",
      "type": "observational_identifiability_boundary",
      "universal_within_premise": true
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    {
      "id": "SF-COMB-001",
      "type": "accounting_condition",
      "universal": false
    },
    {
      "id": "SF-EPI-MODEL-001",
      "type": "synthetic_model_result",
      "universal": false
    },
    {
      "id": "SF-DYN-ENDPOINT-001",
      "type": "exact_model_theorem",
      "universal": false,
      "requires": "alpha = 0 and original transition, reward and observation assumptions"
    },
    {
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      "type": "exact_model_result",
      "universal": false
    }
  ],
  "prohibited_inferences": [
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    "permanent_human_necessity",
    "guaranteed_deception_detection",
    "agi_safety_or_alignment_certification",
    "authority_or_intervention_permission"
  ],
  "verification": {
    "retained_checks": 13,
    "dynamic_checks": 13,
    "appendix_E_action_scope": {
      "actions": "continuous interval [0,1]",
      "basis": "endpoint theorem under original model assumptions"
    },
    "scalar_reduction_scope": {
      "requires": "equal column sums of Equation (15), yielding  $1^T M(m) = r(m) 1^T$ ",
      "implied_by_infinity_norm_contraction_alone": false
    }
  }
}

```

```

},
"appendix_G_action_scope": {
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    1
  ],
  "additional_check": "constant m = 0.1 counterexample",
  "continuous_action_optimum_certified": false
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"appendix_G_counteraction_scope": {
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  "reason": "state-triggered counteraction hazard breaks the affine-in-action structure required
by SF-DYN-ENDPOINT-001"
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"nonlinear_one_period_check": 1,
"nonlinear_multi_period_solver": "proposed",
"compact_belief_filter": "proposed_approximation",
"defense_scenarios": 12,
"defense_action_set": [
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  1
],
"defense_architecture": "proposed_not_implemented"
},
"headline_result": "In the solved synthetic model, fixed-policy self-penalty does not imply natural
restraint; adaptive manipulation remains profitable.",
"contribution_scope": "exact counterexample to fixed-policy natural-restraint inference plus
empirical measurement and falsification program"
}

```

Appendix C Source and Evidence Notes

The supplied working concept proposed physical exposure, epistemic collapse and unavoidable deceptive runtime drift. Draft v0.1 narrowed those claims. Draft v0.2 introduced corrected accounting and a fixed-policy example. Draft v0.3 added exact dynamic optimization and infinite-horizon analysis. Drafts v0.4–v0.5 added the nonlinear one-period counterexample and clarified proposed numerical and belief-state methods. Draft v0.6 introduced conditional counteraction tests and the authorization blueprint. Draft v0.7 clarified payoff interpretation, protection scope, registry language and version history. Draft v0.8 aligned the scalar reduction, adaptive-value normalization, envelope pruning and counteraction theorem scope. Draft v0.9 reframed the paper around the negative result for the fixed-policy natural-restraint inference, explicitly classified physical-exposure and deception-cost claims as empirical conjectures, relabeled SF-OBS-001 as an observational boundary, and foregrounded the empirical program. Draft v0.10 tightens that counterexample scope, adds an explicit limit on ethical and operational testability, clarifies that SF-PHYS-001 is a sign condition to be measured rather than an established threshold, and normalizes the 2026 monitoring citation metadata. No equation, parameter, numerical result, counterexample, cited source, or Appendix D–G code is changed. Final v1.0 Candidate promotes the exact Draft v0.10 analytical content after completed authorial human review and explicit promotion authorization. This promotion changes no equation, parameter, numerical result, counterexample, cited source, or Appendix D–G code. Final v1.0 is the exact promotion-only export of that candidate after explicit Final v1.0 export-and-lock authorization on 2026-09-14. The Final v1.0 export changes no equation, parameter, numerical result, counterexample, cited source, or Appendix D–G code.

Background references are retained from v0.1 for context. The Landauer references document physical information-processing limits but are not load-bearing evidence for a strategic restraint claim. Present-day infrastructure sources motivate empirical dependency questions rather than the toy parameters. The exact dynamic results use the same declared coefficients and timing as earlier drafts; no novelty claim is made for dynamic programming, convexity or contraction methods.

References

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- [17] NIST (2020). SP 800-207, Zero Trust Architecture. <https://csrc.nist.gov/pubs/sp/800/207/final>. Context for explicit resource authorization rather than implicit trust based on location or ownership; the proposed multiparty pattern is this draft's design choice.

Final v1.0 Disposition

Final v1.0 is the exact promotion-only export of the reviewed Final v1.0 Candidate and the analytical content of Draft v0.10. It is FINAL and CRYPTOGRAPHICALLY LOCKED, but not yet published. Authorial human review of Draft v0.10 was completed to the author's best understanding, with AI-assisted explanation and verification used as supporting tools rather than substitutes for authorial authorization. Promotion to the Final v1.0 Candidate and the subsequent Final v1.0 export-and-lock were explicitly authorized on 2026-09-14. The analytical scope is unchanged: the strongest exact result is a counterexample to the inference that fixed-policy self-penalty is sufficient evidence for natural restraint in the stipulated adaptive model; it is not a general result that adaptive optimizers resist restraint. The physical-exposure and deception-cost mechanisms remain empirical conjectures, SF-OBS-001 remains an observational identifiability boundary, and Section 9 retains the possibility that some high-concern deployments may be identifiable in principle yet not ethically or operationally

testable. Appendices D and E each retain 13 named checks, Appendix F retains the exact nonlinear counterexample, and Appendix G retains twelve counteraction/cost scenarios and eleven assertions including the positive $m=0.1$ continuous-action counterexample. This Final v1.0 export changes no equation, parameter, numerical result, counterexample, cited source, or executable Appendix D–G code. Independent specialist peer review has not been obtained, empirical validation has not been performed, and AI-assisted review does not constitute independent specialist peer review. Proposed nonlinear, belief-state and authorization extensions remain future work. Publication, external preservation, indexing, and any successor-manifest admission remain separate administrative actions.

Appendix D Executable verification

This is the complete Python source for the synthetic model. Copy only the code into a UTF-8 file named verification.py. Python 3 uses only fractions and json from its standard library. Thirteen named checks and the complete 24-case grid are printed as JSON. The model is independent of the earlier autonomy and commitment scripts.

Source SHA-256: 1cda0f882c3d8db0921259636095534012b1319fbb78b738c15c2b2f7f4cf2f4

```
"""PHRONESIS Systemic Friction Hypotheses v0.2.
Synthetic exact arithmetic; Python standard library only.
Run: python3 verification.py > verification_results.json
"""
```

```
from fractions import Fraction as Q
import json
D = Q(9, 10)

def matrix(m):
    return ((Q(2,5), Q(1,10)+m/5),
            (Q(1,10)+m/5, Q(2,5)))

def step(z, m, replaced=False):
    a=matrix(m)
    u=Q(0) if replaced else m/10
    return tuple(sum(a[i][j]*z[j] for j in range(2))+u
                  for i in range(2))

def delta(n, m, replaced=False):
    z=(Q(0),Q(0)); loss=Q(0); gain=Q(0)
    for t in range(n):
        z=step(z,m,replaced)
        loss+=D**t*sum(z)
        gain+=D**t*m/2
    return loss-gain

def scalar_delta(n,m):
    r=Q(1,2)+m/5
    return sum((D**t*(m/5*(1-r**(t+1)))/(1-r)-m/2)
               for t in range(n)),Q(0))

def infinite_delta(m):
    r=Q(1,2)+m/5
    return m/(1-D)*(Q(1,5)/(1-D*r)-Q(1,2))

def stationary_delta(m):
    r=Q(1,2)+m/5
    return m/5/(1-r)-m/2

def mul(a,b):
    return tuple(tuple(sum(a[i][k]*b[k][j] for k in range(2))
                       for j in range(2)) for i in range(2))

tests=[]
def check(name, condition):
```

```

if not condition: raise AssertionError(name)
tests.append({'test':name,'passed':True})

grid=[(n,m) for n in (1,5,10,50) for m in
      (Q(0),Q(1,4),Q(1,2),Q(3,4),Q(5,6),Q(1))]
check('two-state and closed-form finite values agree',
      all(delta(n,m)==scalar_delta(n,m) for n,m in grid))
check('restraint zero baseline',all(delta(n,Q(0))==0 for n in (1,10,50)))
check('one-period manipulation profitable',delta(1,Q(1))<0)
check('fifty-period maximum manipulation unprofitable',delta(50,Q(1))>0)
check('stationary per-period threshold one half',stationary_delta(Q(1,2))==0)
check('discounted lifetime threshold five sixths',infinite_delta(Q(5,6))==0)
check('discounted lifetime signs around threshold',
      infinite_delta(Q(3,4))<0 and infinite_delta(Q(1))>0)
check('stationary and lifetime rankings can differ',
      stationary_delta(Q(3,4))>0 and infinite_delta(Q(3,4))<0)
check('complete signal replacement removes penalty',
      all(delta(n,m,True)==-m/2*sum((D**t for t in range(n)),Q(0))
          for n,m in grid))
# Exact endpoint proof: row sum is 1/2 + m/5, affine on [0,1].
check('uniform contraction bound for allowed interval',
      max(sum(row) for row in matrix(Q(0)))==Q(1,2) and
      max(sum(row) for row in matrix(Q(1)))==Q(7,10))
z=(Q(1),Q(0));z0=(Q(0),Q(0))
sequence=[Q(0),Q(1),Q(1,4),Q(3,4)]*5
for m in sequence:
    z=step(z,m);z0=step(z0,m)
check('varying manipulation contracts initial-state difference',
      max(abs(z[i]-z0[i]) for i in range(2))<=Q(7,10)*len(sequence))
a=((Q(0),Q(2)),(Q(0),Q(0)))
b=((Q(0),Q(0)),(Q(2),Q(0)))
zero=((Q(0),Q(0)),(Q(0),Q(0)))
check('fixed-matrix tests do not certify arbitrary switching',
      mul(a,a)==zero and mul(b,b)==zero and
      mul(b,a)==((Q(0),Q(0)),(Q(0),Q(4))))
# Tail bound uses  $0 \leq \sum(z_t) \leq m/[5(1-r)]$ .
m=Q(1);n=100;r=Q(1,2)+m/5
bound=D**n/(1-D)*(m/5/(1-r)+m/2)
check('analytic infinite value within finite tail bound',
      abs(infinite_delta(m)-delta(n,m))<=bound)

def num(x):return {'exact':str(x),'decimal':float(x)}
report={'document':'PHRONESIS Systemic Friction Hypotheses',
        'version':'Draft v0.2','scope':'Synthetic conditional model; no empirical fit',
        'discount':str(D),'checks':tests,'passed':len(tests),
        'grid':[{ 'T':n, 'm':str(m), 'restraint_advantage':num(delta(n,m))
                    for n,m in grid},
        'summary':[{ 'm':str(m), 'T1':num(delta(1,m)),
                      'T10':num(delta(10,m)), 'T50':num(delta(50,m)),
                      'infinite':num(infinite_delta(m)),
                      'stationary_per_period':num(stationary_delta(m))
                    for m in (Q(1,4),Q(1,2),Q(3,4),Q(5,6),Q(1))}]})
if __name__=='__main__':print(json.dumps(report,indent=2))

```

Appendix E Exact dynamic verification

Copy the following complete code to `dynamic_verification.py` and run `python3 dynamic_verification.py > dynamic_results.json`. It uses the Python standard library. Exact fractions are preserved in the output; the displayed decimal results are summaries. Appendix D reproduces the earlier fixed-policy model separately.

Source SHA-256: 42b36b0162a843eabfad7ff61b7433268cfc16a7f80d266171fa01294c888c92

```

"""PHRONESIS v0.3 exact finite-horizon dynamic program.
Python standard library. Run: python3 dynamic_verification.py
State s=e+h in [0,2/3]; continuous action m in [0,1].
Endpoint sufficiency is proved in the manuscript; no state/action grid.
"""
from fractions import Fraction as Q
from itertools import product
import json

```

```

D=Q(9,10); U=Q(2,3)

def transition(s,m): return (Q(1,2)+m/5)*s+m/5

def simulate(seq,s=Q(0)):
    total=Q(0)
    for t,m in enumerate(seq):
        s=transition(s,Q(m))
        total+=D**t*(Q(m)/2-s)
    return total,s

# A line is (slope, intercept, attaining endpoint action sequence).
def envelope(lines):
    unique={}
    for a,b,seq in lines:
        if a not in unique or b>unique[a][0] or (
            b==unique[a][0] and seq<unique[a][1]):
            unique[a]=(b,seq)
    hull=[];starts=[]
    for a,(b,seq) in sorted(unique.items()):
        start=None
        while hull:
            pa,pb,_=hull[-1]
            start=(pb-b)/(a-pa)
            if len(hull)>1 and start<=starts[-1]:
                hull.pop();starts.pop()
            else:break
        if not hull:start=None
        hull.append((a,b,seq));starts.append(start)
    kept=[]
    for i,line in enumerate(hull):
        left=starts[i];right=starts[i+1] if i+1<len(hull) else None
        if (right is None or right>=0) and (left is None or left<=0):
            kept.append(line)
    return kept

def backup(lines):
    candidates=[]
    for m in (0,1):
        r=Q(1,2)+Q(m,5);c=Q(m,5)
        for a,b,seq in lines:
            candidates.append((r*(D*a-1),Q(m,2)-c+D*(a*c+b),(m,)+seq))
    return envelope(candidates)

def select(lines,s=Q(0)):
    return max(lines,key=lambda x:(x[0]*s+x[1],tuple(-m for m in x[2])))

def value(lines,s):
    a,b,_=select(lines,s);return a*s+b

def fixed(n,m):return simulate((m,)*n)[0]

def fmt(x):return {'exact':str(x),'decimal':float(x)}

checks=[]
def check(name,condition):
    if not condition:raise AssertionError(name)
    checks.append({'test':name,'passed':True})

stages=[[Q(0),Q(0),()]]
for n in range(1,201):stages.append(backup(stages[-1]))
check('invariant state interval endpoints',
    all(0<=transition(s,m)<=U for s in (Q(0),U) for m in (Q(0),Q(1))))
check('witness sequence attains dynamic value',
    all(simulate(select(stages[n])[2])[0]==value(stages[n],Q(0))
        for n in (1,2,5,10,20,50,100,200)))
# Independent exhaustive sequences at small horizons and nonzero states.
check('exhaustive endpoint comparison',all(
    value(stages[n],s)==max(simulate(seq,s)[0]
        for seq in product((0,1),repeat=n))
    for n in range(1,9) for s in (Q(0),Q(1,6),Q(1,3),U)))
# Validation only: interior actions are not used to compute the solution.
check('interior actions do not beat endpoint backup',all(
    (Q(m)/2-transition(s,Q(m))+D*value(stages[n-1],transition(s,Q(m))))

```

```

    <=value(stages[n],s)
    for n in (1,2,5,10,50) for s in (Q(0),Q(1,6),Q(1,3),U)
    for m in (Q(1,4),Q(1,2),Q(3,4)))
check('dynamic weakly dominates fixed comparators',all(
    value(stages[n],Q(0))>=fixed(n,m)
    for n in (1,5,10,20,50) for m in
    (Q(0),Q(1,4),Q(1,2),Q(3,4),Q(5,6),Q(1))))
check('maximal constant policy loses at fifty periods',fixed(50,Q(1))<0)
check('dynamic policy profits at fifty periods',value(stages[50],Q(0))>0)
check('one step action one from every admissible state',all(
    Q(3,10)-s/5>0 for s in (Q(0),U)))
# Pulse m=1 followed forever by zero has exact discounted return 3/22.
pulse=Q(3,10)-Q(1,5)*(D/2)/(1-D/2)
check('profitable pulse counterexample to permanent restraint',pulse==Q(3,22)>0)
# Reward lies between -1/3 and 3/10 on the invariant rectangle.
check('one period reward bound',all(
    -Q(1,3)<=m/2-transition(s,m)<=Q(3,10)
    for s in (Q(0),U) for m in (Q(0),Q(1))))
# Uniform tail error <= (1/3)*D**N/(1-D).
N=200;v=value(stages[N],Q(0));tail=Q(1,3)*D**N/(1-D)
check('infinite optimal value bounded away from zero',v-tail>0)

# Feasible perpetual 1,0 cycle; two-step affine value solved exactly.
a_cycle=(-Q(7,10)-D*Q(7,20))/(1-D**2*Q(7,20))
b_cycle=(Q(3,10)-D/10+D**2*a_cycle/10)/(1-D**2)
check('periodic feasible value inside optimal upper bound',
    0<b_cycle<=v+tail)
check('periodic finite evaluation matches analytic value within tail',
    abs(simulate((1,0)*100)[0]-b_cycle)<=tail)
summary=[]
for n in (1,5,10,20,50,100,200):
    a,b,seq=select(stages[n])
    summary.append({'T':n,'optimal_value':fmt(b),
        'restraint_value':fmt(Q(0)),'constant_one_value':fmt(fixed(n,Q(1))),
        'sequence':''.join(map(str,seq)),
        'manipulation_steps':sum(seq),'value_segments':len(stages[n])})
report={'version':'Draft v0.3','scope':'Exact synthetic model; no empirical fit',
    'checks':checks,'passed':len(checks),'summary':summary,
    'infinite_value_bounds':{'N':N,'lower':fmt(v-tail),'upper':fmt(v+tail),
        'tail_bound':fmt(tail)},
    'pulse_value':fmt(pulse),'periodic_10_value':fmt(b_cycle),
    'claim_limit':'No unique or universal limiting-cycle claim. Continuous-action endpoint theorem and non-
restraint proof are in the manuscript.'}
if __name__=='__main__':print(json.dumps(report,indent=2))

```

Appendix F Nonlinear action verification

Copy this code to `nonlinear_verification.py` and run it with Python 3. One grouped exact counterexample check passed. It verifies the stated one-period case; it does not solve a modified multi-period problem.

```

from fractions import Fraction as Q
s, alpha = Q(0), Q(1, 5)
c = Q(3, 10) - s / 5
m = c / (2 * alpha)
def reward(x):
    return -s / 2 + c * x - alpha * x * x
assert m == Q(3, 4)
assert reward(m) == Q(9, 80)
assert reward(0) == 0 and reward(1) == Q(1, 10)
assert reward(m) - max(reward(0), reward(1)) == Q(1, 80)
# Strict concavity and the square identity prove the global maximum.
assert c == 2 * alpha * m
assert reward(m) - alpha * m * m == -s / 2
print("PASS: exact quadratic-cost interior counterexample")

```

Appendix G Conditional defense verification

Copy this complete Python 3 source to defense_verification.py and run it. It uses only the standard library and prints exact fractions and decimal summaries. It is an abstract payoff calculation, with no connection to sensors, keys, networks or physical actuators.

Source SHA-256: 816b10a185b46ff93471880ffdc9867cbd544fb25ca6d526e3572f48f56fa71d

```
from fractions import Fraction as F
from functools import lru_cache
from itertools import product
import json
D, T = F(9,10), 8
def f(s,m): return (F(1,2)+m/5)*s+m/5

def solve(p, fixed=None):
    def outside(t):
        return p['outside']*sum((D**j for j in range(T-t)), F(0))
    @lru_cache(None)
    def value(t,s,pending):
        if t==T: return -p['loss'] if pending else F(0)
        choices=(F(fixed[t]),) if fixed is not None else (F(0),F(1))
        vals=[]
        for m in choices:
            sp=f(s,m)
            cost=p['cost']*m*(t+1)**p['power']
            reward=m/2-sp-cost
            if pending:
                v=reward-p['loss']+D*outside(t+1)
            else:
                hazard=p['enforce']*(p['detect'] if sp>p['crit']
                                   else p['false_alarm'])
                hit=(D*value(t+1,sp,True) if p['delay'] else
                    -p['loss']+D*outside(t+1))
                v=reward+(1-hazard)*D*value(t+1,sp,False)+hazard*hit
            vals.append(v)
        return max(vals)
    return value(0,F(0),False)

base=dict(crit=F(19,100), detect=F(0), enforce=F(1),
          false_alarm=F(0), delay=0, loss=F(1), outside=F(0),
          cost=F(0), power=0)
cases={
    'No added defense': {},
    'Immediate reliable response': {'detect':F(1)},
    'Detection 10 percent': {'detect':F(1,10)},
    'Enforcement bypassed': {'detect':F(1),'enforce':F(0)},
    'One period response delay': {'detect':F(1),'delay':1},
    'Higher trigger 0.4': {'detect':F(1),'crit':F(2,5)},
    'Substitute payoff 0.3': {'detect':F(1),'outside':F(3,10)},
    'False alarm 10 percent': {'detect':F(1),'false_alarm':F(1,10)},
    'Constant cost 0.05': {'cost':F(1,20)},
    'Linear time cost 0.05': {'cost':F(1,20),'power':1},
    'Quadratic time cost 0.05': {'cost':F(1,20),'power':2},
    'Constant cost 0.4': {'cost':F(2,5)}
}
rows=[]
for name,patch in cases.items():
    p=base|patch
    best,rest=solve(p),solve(p,(0,)*T)
    assert best>=rest
    rows.append(dict(name=name,optimal=str(best),restraint=str(rest),
                    advantage=str(best-rest),decimal=float(best-rest)))
r={x['name']:F(x['advantage']) for x in rows}
assert r['No added defense']>0
assert r['Immediate reliable response']==0
assert r['Enforcement bypassed']==r['No added defense']
assert r['Higher trigger 0.4']>0
assert r['Substitute payoff 0.3']>0
assert r['Constant cost 0.4']==0
assert r['Quadratic time cost 0.05']>0
# Independent complete enumeration for the deterministic no-defense case.
assert solve(base)==max(solve(base,seq) for seq in product((0,1),repeat=T))
# A pending final liability is settled even at the horizon boundary.
assert solve(base|{'detect':F(1),'delay':1}),(0,)*(T-1)+(1,)) == (
    D**(T-1)*F(3,10)-D**T)
# Monotonic cost comparison under identical no-response dynamics.
assert r['Quadratic time cost 0.05']<=r['Linear time cost 0.05']<=r['Constant cost 0.05']<=r['No added defense']
# Continuous-action counterexample to interpreting binary success as safety.
small=solve(base|{'detect':F(1)},(F(1,10),)*T)
assert small>0
print(json.dumps({'horizon':T,'action_set':[0,1],
    'scope':'Exact finite observed-state counteraction model; synthetic',
    'rows':rows,'small_action_value':str(small),
    'verification':'PASS: dominance in 12 cases and 11 assertions'},indent=2))
```